Lecture Note on Vector

Scalars and Vectors

Scalar

A **scalar** is a physical quantity that has **magnitude only** but no direction.
Examples: Mass (5 kg), Temperature (30°C), Time (2 s), Speed (20 m/s).

Vector

A **vector** is a physical quantity that has both **magnitude and direction**.
It is usually represented by an arrow: the *length* shows magnitude, and the *arrowhead* shows direction.
Examples: Displacement (5 km east), Velocity (20 m/s north), Force (10 N downward), Acceleration.

Representation of a Vector

- **Graphical Representation**: As an arrow drawn to scale.

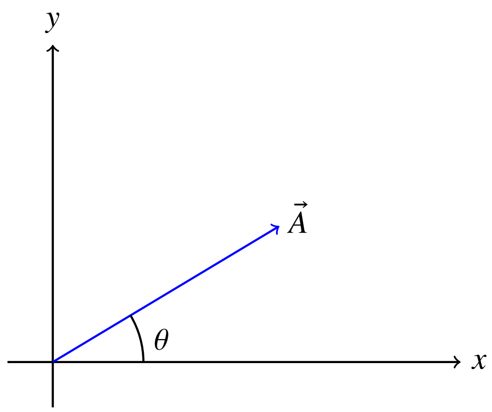

Figure 1: Graphical representation of a vector $\vec{A}$ in the xy-plane.

- **Analytical Representation using Unit Vectors**: A unit vector has magnitude 1 and points in a given direction.

$$\hat{i}\ \text{(x-axis)}, \quad \hat{j}\ \text{(y-axis)}, \quad \hat{k}\ \text{(z-axis)}$$

Any vector in 3D can be expressed as

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}.$$

Components of a Vector

If $\vec{A}$ makes an angle θ with the x-axis in 2D:

$$A_x = A\cos\theta, \quad A_y = A\sin\theta$$

Thus,

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath}$$

Magnitude:

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}, \qquad \tan\theta = \frac{A_y}{A_x}$$

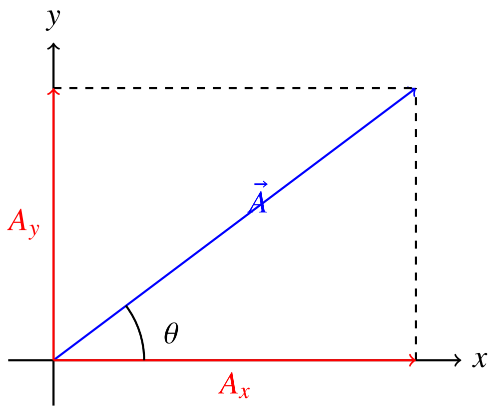

Figure 2: Vector $\vec{A}$ and its components along x and y axes.

Vector Algebra

Addition of Vectors:

If

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath}, \qquad \vec{B} = B_x\hat{\imath} + B_y\hat{\jmath},$$

then

$$\vec{R} = \vec{A} + \vec{B} = (A_x + B_x)\hat{\imath} + (A_y + B_y)\hat{\jmath}.$$

Subtraction of Vectors:

$$\vec{A} - \vec{B} = (A_x - B_x)\hat{\imath} + (A_y - B_y)\hat{\jmath}.$$

Multiplication of Vectors:

Scalar (Dot) Product

$$\vec{A}\cdot\vec{B} = |\vec{A}||\vec{B}|\cos\theta$$

Result is a scalar. Example: Work done = $\vec{F}\cdot\vec{d}$.

Vector (Cross) Product

$$\vec{A} \times \vec{B} = |\vec{A}||\vec{B}| \sin\theta \, \hat{n}$$

Result is a vector perpendicular to both $\vec{A}$ and $\vec{B}$. Example: Torque = $\vec{r} \times \vec{F}$.

Example

Worked Example

A car travels 10 km east and 5 km north. Represent the displacement vector.

$$\vec{R} = 10\hat{i} + 5\hat{j} \text{ (km)}$$

Magnitude:

$$|\vec{R}| = \sqrt{10^2 + 5^2} = \sqrt{125} = 11.18 \, \text{km}$$

Direction:

$$\theta = \tan^{-1}\left(\frac{5}{10}\right) = 26.6^\circ \text{ north of east.}$$